beyond classical PDEs
conservation laws shocks, entropy solutions
Hamilton–Jacobi viscosity solutions
nonlocal & fractional operators
integrable systems & solitons
optimal transport & gradient flows
memory & viscoelastic transport
microcontinuum continua
nonlinear diffusion porous medium
delay & retarded transport
Cattaneo & telegraph finite-speed diffusion
variational inequalities hysteresis
variational principles & Noether
homogenisation & upscaling
stochastic PDEs random media
free boundaries level sets, phase field
MHD & relativistic transport
numerics & inversion
structure-preserving discretisations
inverse problems & data assimilation
machine learning & operator learning
nonlinear solvers continuation
linear solvers preconditioning
adaptivity, moving meshes parallel decomposition
particle & Monte Carlo hybrid methods
CFL, stiffness time stepping
operator splitting & multiphysics coupling
discretisation: consistency stability, convergence
verification & validation
numerical diffusion & dispersion
limiters · TVD · WENO
bridges to other fields
finance
images & signals
networks & traffic
machine learning & data
one mathematics of transport
quantum & condensed matter
geoscience & climate
biology & medicine
plasma & astrophysics
engineering devices
foundations
PDE taxonomy + boundary conditions
higher order & dispersive
entropy & variational structure
dimensions, geometry & manifolds
Fourier, Green functions propagators
material derivative Lagrangian ↔ Eulerian
memory & linear response Mori–Zwanzig
dimensionless numbers Pe · Re · Ra · Da · Kn
the balance law flux · divergence
constitutive laws Fick · Fourier · Darcy
random walks, Langevin fluctuation–dissipation
bulk fluid transport
non-Newtonian rheology
vorticity & pressure
shear dispersion Taylor–Aris
Darcy & porous flow
convection & buoyancy Rayleigh–Bénard, Marangoni
turbulence & eddy diffusivity
jets · plumes · wakes
boundary layers
advection
mixing & chaotic advection
recurring mechanisms — the shapes that keep returning
nonlinearity & mode coupling
scaling & self-similarity
duality & transforms
equilibria & critical points
linearisation & stability
bifurcation theory
chaos & predictability
phase transitions & criticality
noise-induced phenomena
the intersections map
bibliography (primary · review · sceptical)
kinetic & radiative
radiative transfer
neutron transport
Boltzmann equation H-theorem
lattice-Boltzmann particle methods
Fokker–Planck Kramers escape
BGK · Vlasov plasmas
interaction / source / sink
accretion · ablation surface growth
autocatalysis & oscillations
reaction–diffusion Turing patterns
chemotaxis Keller–Segel
subsidence & poroelasticity
wells · injection · efflux breakthrough
phase change Stefan problem
fronts & travelling waves
absorption · dissolution interfacial transfer
reaction kinetics Arrhenius
viscous dissipation Joule heating
sedimentation · deposition filtration
sorption & retardation
microscopic transport
conduction (Fourier)
diffusion coefficients from kinetic theory
Knudsen & rarefied transport
diffusion (Fick)
Nernst–Planck & electrochemical
Soret · Dufour Maxwell–Stefan
anomalous diffusion CTRW, fractional
Brownian motion Einstein relations
universality — the strongest claim
the atlas (master note)
physics, chemistry & field theory
mathematics itself (geometric flows, categories)
why everything flows (five universal structures)
the translation lookup (object → transport form)
computation & algorithms (Dijkstra, PageRank, SGD)
limits of the claim (where it stops)
information, statistics & inference
cognitive science & mind (decisions, fields, memory)
social behaviour, economics & language
panta rhei — all things flow
collective behaviour & synchrony
crossings & cross-fertilisation
the river as a model
beauty & the flow (marbling, patterns)
Heraclitus & the river
irreversibility & the arrow of time